DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
West Point, New York 10996

MADN-MTH 2 November 2026

MEMORANDUM FOR DXDT "Velocity" Scholarship Committee, United States Military Academy, West Point, NY 10996

SUBJECT: Application for AY 2027 DXDT "Velocity" Scholarship

1. Long-term Professional Goals. I intend to serve as an Army officer who turns operational problems into rigorous models and fielded software. I plan to commission as an Infantry officer and lead Soldiers first, because I believe many crucial problems start within formations. As an Operations Research major with a minor in artificial intelligence and a finance specialization, I then want to build toward service as an Operations Research/Systems Analyst (FA49) or work with AI through the Army AI Integration Center (AI2C). I plan to achieve those career paths by earning a master's degree in artificial intelligence engineering at Carnegie Mellon University, through the Army's AI Scholars Program, or at MIT. Most importantly, I want to return to West Point to teach in the Department of Mathematical Sciences, bringing that research and AI expertise into the classroom.

2. Purpose. Mr. Philippone has described data as the ammunition of the future and has framed a commander's hardest choices as constrained optimization, which means finding the best decision across competing trade-offs fast enough to matter. This past June, on the Technology and Innovation AIAD through SOSH, separate from the Sea to Shining Sea Fellowship I am a part of during the academic year, I saw that idea from several angles. Palantir's team in Palo Alto walked us through Project Maven and how it gives commanders at every echelon a common picture to decide from, and the 4th Infantry Division showed us how a formation employs those systems and feeds lessons back to the developers. At Stanford's Gordian Knot Center, COL (Ret.) Joseph Felter and Jack Faherty described how student startups get new technology into the hands of Army units, and why that is so difficult. At Meta, we met with LTC Andrew Bosworth, Meta's chief technology officer and an Army Reserve officer in Detachment 201, which brings technology executives into the Army.

At my second AIAD, with the Army's Transformation Decision Analysis Center (TDAC), I redesigned the reward system of a reinforcement-learning model for contested resupply. I am pursuing the DXDT "Velocity" Scholarship for two reasons. First, its funding would take my research out of the classroom to defense conferences where I can present it, a hackathon where I can compete, and an AIAD where I can learn how similar tools are tested before they reach a commander. Second, as a Velocity Scholar I would seek to work alongside industry partners such as Palantir and Shield AI, whether through an AIAD, a hackathon, or a joint research project, and bring that practice back to my research and my future formation.

MADN-MTH
SUBJECT: Application for AY 2027 DXDT “Velocity” Scholarship

3. Near-term Research Goals. My interest in contested logistics began at TDAC, where I redesigned the reward function for a reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army’s sustainment simulation. I built diagnostic tools that exposed why a high-scoring agent was starving the priority unit, and the retrained agent more than tripled the share of requests filled. That work showed me how a supply system can look successful while failing the unit that needs it most. This experience was fruitful and left me with another research question to pursue. I wanted to figure out how an adversary would strike a resupply route to limit max-flow.

That question became my current project with my advisor, LTC Matthew Webb, whose doctoral research was in transportation and logistics at MIT. My project asks where an adversary with a fixed effort budget should act to cut Taiwan’s crude oil supply. Additionally, it tests whether LTC Webb’s reinforcement-learning attacker, trained on Ukraine’s rail network, still finds strong strategies on a maritime network. I modeled Taiwan’s crude import routes as a 61-node, 83-edge capacitated network, built from open sources and checked against an independent max-flow solver. The adversary has nine types of action, each with an effort cost and a chance of success, and each plan is scored by expected max-flow, the share of demand Taiwan still receives after rerouting averaged over which actions succeed, which the adversary seeks to minimize.

My near-term goal is to finish this study this semester. I will (1) test whether the trained agent beats random plans, which leave Taiwan about 90% of its demand, and a greedy baseline, which leaves 70%, and comes close to an exact solver while answering in seconds rather than hours, (2) run six wargaming scenarios, from the base case to lower and higher action costs, higher and lower demand, and a closed Strait of Hormuz, to see how the adversary’s best plan changes, and (3) report whether a rail-trained attacker transfers to a maritime network.

Over the rest of this academic year, I will extend the work to a traveling salesman problem with neighborhoods, in which a route only has to pass within reach of each target area rather than through a fixed point. Applied to my network, it asks how a small fleet of escort or resupply ships could cover every chokepoint the interdiction model flags in the least time, turning the attacker’s plan into a defender’s route. I will present both results at Projects Day and, if accepted, at SIEDS.

In my remaining two and a half years at West Point, I want to take on harder versions of these problems, such as larger networks, uncertain capacities, and planning across several time periods, and build them into a research line I can present at INFORMS and MORS. As a platoon leader, I will stay connected to the academic world by attending conferences and conventions such as AUSA and MORS, presenting when my duties allow, and keeping the research ties I build here. That keeps my skills current for graduate school and for returning to teach.

4. Funding Plan and Budget.

a. Funding Plan. I plan to attend one AIAD, one conference, complete three online courses, and use one AI research subscription. I list the AIAD and conference options in priority

order and will attend only one of each, so each group carries a single estimate of $6,700 for the AIAD and $2,000 for the conference.

AIAD, Primary. In summer 2027, I will spend $6,700 on housing, food, and transportation for a three-week AIAD with Palantir or Shield AI, the AIADs for which Velocity Scholars receive priority fill. Both companies build the systems that turn models like mine into decisions, Palantir through the Maven Smart System and Shield AI through its autonomy software. Reconnecting with the Palantir team I met in Palo Alto this June would let me see how and why industry is outpacing the Department of Defense in fielding AI.

For me, the AIAD shows how an engineering team turns a mathematical model into software a commander trusts and acts on, from integrating messy data to designing what the user sees, which is what any model I build must survive to matter outside a research paper. For the Department of Mathematical Sciences, I will brief LTC Webb and the DXDT committee within 30 days of returning and write a two-page after-action report the next Velocity cohort can use to prepare for the same placement. For West Point, a cadet who has worked inside Palantir or Shield AI keeps the DXDT partnership active and gives the next cadet a contact and a precedent to build on.

AIAD, Alternate 1. If neither placement is available, I will do a three-week AIAD from 11–31 July 2027 at MIT Lincoln Laboratory in Lexington, MA, a Department of Defense research center that builds and tests AI and decision-support tools for the military. At TDAC, the agent I first worked on scored well while starving the priority unit, a reward-gaming problem. At Lincoln Laboratory I would further my reinforcement-learning skills and apply my reward-design experience to ongoing defense-sponsored projects.

For me, the payoff is learning how professional researchers validate a model before anyone relies on it, through held-out test cases, stress tests, and adversarial checks, a discipline I will carry into every model I build as a cadet and officer. For the Department of Mathematical Sciences, I will turn that process into a one-page testing checklist for cadet reinforcement-learning projects and give it to LTC Webb for the cadets he advises. For West Point, the AIAD strengthens the Academy’s relationship with a Department of Defense laboratory and opens a placement future cadets can request.

Conference, Primary. On 24–25 May 2027, I will spend $2,000 on train travel, lodging, and meals to attend the SCSP AI+ Expo in Washington, DC; admission is free. It ranks first because it is one of the largest gatherings of defense-technology companies and government AI leaders in the country, with more than 18,000 attendees. If SCSP runs its National Security Technology Hackathon again, I will enter with a team of cadets.

For me, the hackathon tests whether I can turn mathematics into a working prototype under time pressure, and the Expo shows me which problems in optimization and AI the defense community most needs solved, which will shape the research I take on in my remaining years at West Point. For the Department of Mathematical Sciences, I will recruit the hackathon team from operations research and mathematics majors and bring back a list of firms and points of contact that take cadets for AIADs or sponsor capstone projects. For West Point, cadets competing in a national-security hackathon show defense-technology leaders what the Academy’s math majors can build.

Conference, Alternate 1. In early May 2027, I will spend $2,000 on travel, lodging, meals, and registration to present my Taiwan interdiction paper at the IEEE Systems and Information Engineering Design Symposium (SIEDS) at the University of Virginia, if accepted. It ranks second because acceptance depends on my paper, but presenting would give the project a peer-reviewed, citable publication in IEEE Xplore.

For me, peer review is the standard every serious researcher meets, and meeting it as a cadet teaches me how to frame a problem, defend a method, and state its limits in a form other researchers can check. For the Department of Mathematical Sciences, it would be a peer-reviewed undergraduate publication from LTC Webb’s line of interdiction research, and a worked example future cadets can follow to take an independent project to SIEDS. For West Point, it puts a cadet’s operations research in front of the national systems engineering community.

Online Courses. During 2027, I will spend $797 on verified certificates for three courses in the MITx Supply Chain Management MicroMasters program, offered by the MIT Center for Transportation & Logistics. The courses are Supply Chain Analytics (SC0x, $199), Supply Chain Fundamentals (SC1x, $299), and Supply Chain Design (SC2x, $299).

Supply Chain Analytics will develop my ability to quantify uncertainty through probability, statistical analysis, hypothesis testing, and regression, strengthening the statistical foundation under every model I build. Supply Chain Fundamentals will deepen my understanding of procurement, inventory, transportation, and distribution, helping me model real supply networks more faithfully. Supply Chain Design covers network design and optimization and teaches how to use optimization and simulation to decide where added capacity or protection best preserves supply under disruption.

Together, these courses will help me move from identifying supply chain vulnerabilities to developing models that help commanders defend against them, and they count toward the MicroMasters credential, a pathway to graduate study in supply chain management. For the Department of Mathematical Sciences, the probability and regression in Supply Chain Analytics overlap with MA103, so I will bring its practice problems into the MA103 tutoring I lead for 1,010 cadets.

AI Research Subscription. In January 2027, I will spend $500 on a Claude Enterprise subscription from Anthropic for the 2027 research year. I will use it as a coding assistant to build, test, and document the optimization and reinforcement-learning code behind my research, so LTC Webb and future cadets can reproduce every result. I will use it only with unclassified, open-source data, never with controlled or classified information. For the Department of Mathematical Sciences, I will write a short guide on using AI coding assistants responsibly in cadet research and share it with cadets doing independent studies.

MADN-MTH
SUBJECT: Application for AY 2027 DXDT “Velocity” Scholarship

b. Research Travel. Options are listed in priority order. I will attend one AIAD and one conference, so each group has one estimated cost.

Purpose of Trip (AIAD, Conference, other)	Destination	City, State	Dates or # of Days	Estimated total Trip costs
AIAD, Primary	Palantir or Shield AI	Per placement	Summer 2027 (21 days)	$6,700
AIAD, Alternate 1	MIT Lincoln Laboratory	Lexington, MA	11–31 Jul 2027	
Conference, Primary	SCSP AI+ Expo and hackathon	Washington, DC	24–25 May 2027	$2,000
Conference, Alternate 1	IEEE SIEDS, UVA	Charlottesville, VA	Early May 2027	

c. Equipment, Materials, or Supplies. Three MITx course certificates from MIT (learn.mit.edu), SC0x ($199), SC1x ($299), and SC2x ($299), for $797, and a Claude Enterprise subscription from Anthropic (anthropic.com) for $500, for a total of $1,297.

d. Total. $9,997 (one AIAD $6,700, one conference $2,000, online courses $797, AI research subscription $500), within the $10,000 limit.

e. Timeline for Expenditures. From November 2026 to February 2027, I will request my AIAD choices in priority order, submit the SIEDS abstract, register for the conferences, and buy the Claude subscription. I will attend the funded conference between March and May 2027 and complete the AIAD in summer 2027. I will buy each MITx certificate when that course opens, and all funds will be spent by 31 December 2027, before my May 2029 graduation.

5. The point of contact for this memorandum is CDT Dennis Pezan at (815) 592-6927 or dennis.pezan@westpoint.edu.

Encl
Applicant’s Curriculum Vitae

DENNIS G. PEZAN
CDT, USCC
Team Leader, Company C-2